

L10a: Monte Carlo Pricing and Sensitivity Analysis of American Options Premium

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced American options on a CRR tree. Can simulation price them? Duration and convexity measured how a bond's price moves with its yield. How much does the premium change when the share price, volatility, time to expiration, or interest rate changes?

Today's concepts

- Estimate and decompose the premium of American options: Longstaff–Schwartz Monte Carlo, then intrinsic value (immediate exercise value) and extrinsic value (time value).
- Compute the Greeks for American options: Delta, Theta, Vega, Rho, and Gamma from a local Taylor expansion of the options premium.
- Interpret Delta and compute the probabilities of exercise and profit: the shares needed to hedge an option position, and the probabilities from projected share price distributions at the strike price and at the breakeven.

Lectures and Examples

Lecture:
CHEME-5660-L10a-Lecture-American-Derivatives-Greeks-Fall-2026.ipynb

Example 1:
CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 3:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Review on your own:
CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

Company Profile: Susquehanna

Susquehanna (SIG) is a privately held trading firm whose core business is options trading. Friends who traded options on the floor of the Philadelphia Stock Exchange started it in 1987.

- Options market making: it posts prices to buy and sell, so it takes the other side of its customers' contracts. In October 2026, it was the Designated Primary Market Maker for 1,359 of 5,370 Cboe option classes, the most of any firm.
- Managing the Greeks: its book is exposed to Delta, Vega, Theta, and Rho at once, and the goal is to net these across products.
- Training: interns learn options theory, and poker teaches new traders to decide under uncertainty.

Connection to today's lecture: when a customer buys a call, the firm hedges with $100\,\Delta_c$ shares and rebalances as Gamma moves Δ_c.

Monte Carlo Pricing of American Options

We simulate share price paths under the risk-neutral pricing rule $\mathbb{Q}$, then estimate the value of holding at each date by regressing the discounted cash flows from holding on the current share price across paths. Like the lattice, Longstaff and Schwartz work backward from expiration:

Initialize: given the share price S_0, the strike price K, the payoff function $V(K, S)$ of the call or put, the time to expiration T (years), the volatility σ, the risk-free rate g_y, the number of exercise dates h after today, and the number of paths M. Compute $\Delta t = T/h$ and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$ (expiration), where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths, those with $V(K, S_i(t)) > 0$, fit the value of holding $\hat{C}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{C}_t(S_i(t))$, **then**: exercise, so $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let's price an American put by simulation and compare it with the lattice and the European premium. We apply two rules to the same simulated paths:

Rule at each date	Estimates
Longstaff–Schwartz: exercise when $V(K, S_i(t)) > \widehat{C}_t(S_i(t))$	$\mathcal{P}^{\text{AM}}$
Hold on every path	$\mathcal{P}^{\text{EU}}$

$$\boxed{\text{early-exercise premium} \approx \mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}}}$$

Example 1: CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Extrinsic versus Intrinsic Value

The premium (price) of an American option can be broken down into two components: intrinsic value and extrinsic value.

- Intrinsic value: the immediate exercise value, $\max(0, S(t) - K)$ for a call and $\max(0, K - S(t))$ for a put, where $S(t)$ is the current share price.
- Extrinsic value: also known as time value, the additional value from the potential for future price movements before expiration.

The extrinsic value $E_V(t)$ can be expressed mathematically as:

$$E_V(t) = \text{Option Price} - \text{Intrinsic Value} \quad \blacksquare$$

Extrinsic Value

In expanded form, the extrinsic value $E_V(t)$ is:

$$E_V(t) = \begin{cases} \mathcal{P}_c(K, S(t)) - \max(0, S(t) - K) & \text{(for Call Options)} \\ \mathcal{P}_p(K, S(t)) - \max(0, K - S(t)) & \text{(for Put Options)} \end{cases}$$

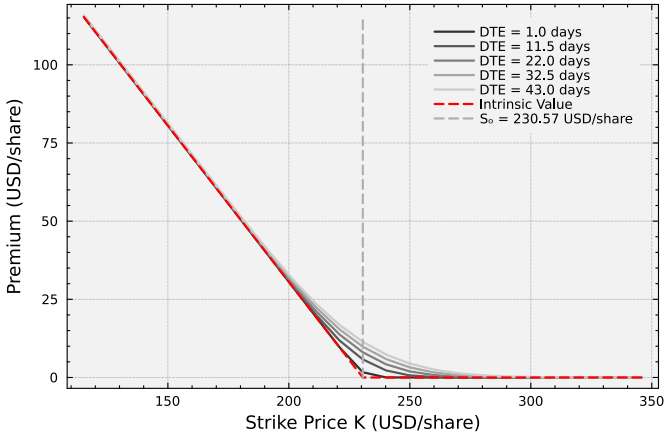
When the option is out of the money, it has no intrinsic value, and the entire option price is extrinsic value.

Why do people pay money for something that is "worthless"? There is still a chance that the share price moves favorably before expiration.

As time moves towards expiration, the extrinsic value tends to decrease (time decay), because there is less time for the share price to move favorably.

Review on your own: CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

Call Premium across Strikes and Expirations



Hmmmm: the premium changes with time to expiration and the strike price. How does it change with other factors?

The Greeks

The Greeks are local sensitivity coefficients that measure how American options premiums change with market conditions:

- Delta Δ: the change in premium for a `+1 USD/share` change in the share price.
- Theta Θ: the change in premium for a `1-day decrease` in the lifetime of the contract (time decay).
- Vega ν: the change in premium for a `+1%` change in volatility.
- Gamma Γ: the change in Δ for a `+1 USD/share` change in the share price.
- Rho ρ: the change in premium for a `+1%` change in the risk-free rate, usually minor except for longer-term options.

Most trading platforms compute these for you!

A Local Taylor Expansion of the Premium

Expanding the premium $\mathcal{P}_\star(S, \sigma, t, g_y)$ of a contract of type $\star \in \{c, p\}$ at calendar time t to second order in S and first order in the other inputs:

$$\boxed{\Delta\mathcal{P}_\star \approx \Delta_\star\,\Delta S + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2}$$

$$\Delta_\star = \frac{\partial \mathcal{P}_\star}{\partial S} \qquad \nu_\star = \frac{\partial \mathcal{P}_\star}{\partial \sigma} \qquad \Theta_\star = \frac{\partial \mathcal{P}_\star}{\partial t} \qquad \rho_\star = \frac{\partial \mathcal{P}_\star}{\partial g_y} \qquad \Gamma_\star = \frac{\partial^2 \mathcal{P}_\star}{\partial S^2}$$

Derivatives are taken at the current inputs, with Δt in years and $\Delta\sigma$, Δg_y in decimals. Platforms quote $\Theta_\star/365$ per day, and $\nu_\star/100$ and $\rho_\star/100$ per percentage point.

Delta and Gamma as Duration and Convexity

$\Delta_\star$ and $\Gamma_\star$ play the roles of a bond's duration and convexity: the slope and the curvature of the premium as a function of the share price.

Example: a long call has $\Delta_c = 0.50$ and $\Gamma_c = 0.02$ per USD/share.

$$\Delta S = +5: \quad \Delta \mathcal{P}_c \approx 0.50 \times 5 + \tfrac{1}{2}(0.02)(5)^2 = 2.50 + 0.25 = 2.75 \text{ USD/share}$$

$$\Delta S = -5: \quad \Delta \mathcal{P}_c \approx -2.50 + 0.25 = -2.25 \text{ USD/share}$$

The Gamma term adds to the premium whichever way the share price moves.

Computing the Greeks of American Options

Backward induction on the CRR tree gives no formula to differentiate, so we reprice the tree with one input moved up and down by a step ϵ:

$$\frac{\partial \mathcal{P}_\star}{\partial x} \approx \frac{\mathcal{P}_\star(x+\epsilon) - \mathcal{P}_\star(x-\epsilon)}{2\epsilon} \qquad x \in \{S, \sigma, t, g_y\}$$

$$\Gamma_\star \approx \frac{\mathcal{P}_\star(S+\epsilon) - 2\,\mathcal{P}_\star(S) + \mathcal{P}_\star(S-\epsilon)}{\epsilon^2}$$

With no dividends, a call's American and European premiums are equal, so the Black–Scholes–Merton formulas give Greeks to check against, e.g., $\Delta_c = N(d_+)$.

Example 2: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

The Secret Lives of Delta

Besides measuring the premium change for a `+1 USD/share` move in the share price, Δ has two other interpretations:

- Hedge ratio: for every n contracts held ($n < 0$ if the contracts are sold), hold $-100 \cdot n \cdot \Delta$ shares to be delta-neutral, i.e., insensitive to small share price moves.
- Probability of exercise: Δ is often used as an estimate of the probability that the option expires in-the-money. A call with $\Delta = 0.7$ suggests a 70% chance.

Probability of Exercise (POE)

The probability that the contract finishes in-the-money at expiration $t = T$:

$$\text{POE} = \begin{cases} \mathbb{P}\left(S_T > K + 0.01\right) & \text{(for Call Options)} \\ \mathbb{P}\left(S_T < K - 0.01\right) & \text{(for Put Options)} \end{cases}$$

where S_T is the share price at expiration. In the Black–Scholes–Merton model, a European call has $\Delta_c = N(d_+)$, while the risk-neutral probability that it finishes in-the-money is $N(d_-)$, which is smaller.

Conjecture: $\Delta \approx$ POE. Example 3 checks it.

Probability of Profit (POP)

The probability that the contract's profit at expiration is positive:

$$\boxed{\text{POP} = \mathbb{P}\left(\hat{P}_{\mathcal{C}} > 0\right)}$$

where $\hat{P}_{\mathcal{C}}$ is the profit of the (single or composite) contract at expiration.

POE reads the projected distribution at the strike price, POP at the breakeven.

Example 3: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Optional Advanced Material

Numerical Greeks: advanced/numerical_greeks/
CHEME-5660-L10a-Advanced-Numerical-Greeks-Fall-2026.ipynb

How accurate are Greeks computed by repricing a binomial tree? The CRR premium bends only where a terminal node crosses the strike, so a step that stays between two bends gives a Gamma of zero. The Leisen–Reimer tree keeps the call Greeks close to the Black–Scholes–Merton values at every number of tree steps we try. We then plot an American put's Delta and Gamma around the early-exercise boundary.

Notebooks: lectures/week-10/L10a/advanced, also linked from the lecture notebook.

Summary

Today we priced American options by simulation, split the premium into intrinsic and extrinsic value, and introduced the Greeks.

Key takeaways

- Estimating and decomposing the American premium: Longstaff–Schwartz regresses the value of holding on the share price. Intrinsic value is the immediate exercise value, and extrinsic value is the potential for favorable price movement.
- The Greeks quantify option price sensitivity: they are the partial derivatives in a local Taylor expansion of the premium, which we estimate for American options by repricing the CRR tree.
- Delta serves three purposes: the rate of change in the premium, the hedge ratio for delta-neutral positions, and an approximation of the probability of exercise. POE reads the projected distribution at the strike price, and POP reads it at the breakeven.